BOOKNEST PROJECT DOCUMENTATION

Digital Egypt Pioneers' Initiative (DEPI)

BookNest

ONLINE LIBRARY

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1. Project Planning & Management

1.1 Project Proposal

Overview:

BookNest is a comprehensive Android-based digital library designed to simplify book discovery and reading. It provides a unified platform where users can explore various genres, read books internally (In the future), and manage their personal collections efficiently.

Objectives:

- Centralized Library Platform: To create a unified and organized digital catalog where books are classified into multiple categories for easy navigation.
- Seamless In-App Reading Experience: To implement in the future a high-performance internal PDF viewer that allows users to read books directly within the application.
- Offline Accessibility: To provide a robust download mechanism in the future that enables users to access their favorite books without an internet connection.
- User Personalization: To build an engagement system including "Search" and "Favorites" features for tailored user experience.
- Performance & Scalability: To ensure a fast and stable application using Kotlin and MVVM architecture.

Scope:

The project includes User Authentication, Category Management, Favorites , and Search functionality.
Note: To ensure security and reduce technical complexity at this stage, BookNest **will not include an internal payment gateway**. Instead, it will provide direct external links for purchasing premium books to mitigate financial transaction risks.

1.2 Project Plan (Timeline & Milestones)

Week 1 – Project Setup & Core UI :

Goals:

- Set up project structure in Android Studio.
- Configure Gradle dependencies (Jetpack Compose, Retrofit, Coroutines, Coil/Glide, etc.).
- Define app architecture (Clean Architecture).
- Create mock data to design UI without API.
- Build core screens with Compose:
    - Home Screen (list of books)
    - Book Details Screen (basic layout)
    - Navigation setup with Navigation-Compose.

Deliverable: *Functional app with navigation and mock UI for book list & details.*

Week 2 – API Integration & Data Layer

Goals:

- Connect to the chosen API.
- Implement Retrofit for network calls.
- Set up data layer (Repositories).
- Use Coroutines + Flow for async data.
- Replace mock data with real API responses.
- Implement error handling & loading states.

Deliverable: *Book list fetched from API and displayed in the Home Screen.*

Week 3 – Features & State Management

Goals:

• Implement Category Filtering (Group books by genre).

• Implement search functionality (search book by name).

• Improve UI with Material3 design.

• Apply proper state handling (e.g., using ViewModel).

Deliverable: *Searchable and Categorized book list with polished UI.*

Week 4 – Testing, Polish & Deployment

Goals:

• Unit tests for ViewModel & Repository.

• UI tests with Compose Testing.

• Optimize performance (lazy lists, image loading).

• Improve UX (error dialogs).

• Add splash screen & app icon.

• Prepare Play Store-ready build (if needed).

Deliverable: *Fully functional, tested, and polished Books Library app.*

1.3 Task Assignment & Roles

To ensure a high-quality delivery for the **BookNest** project, tasks have been distributed based on the app's modules and technical requirements:

Ahmed Sameh (Team Leader & Authentication Developer):

- **Project Architecture: Designed the MVVM architecture and managed project dependencies.**
- **Version Control: Managed the GitHub repository, branches, and code integration.**
- **Authentication: Implemented the complete authentication flow, including Login, Sign Up, and user session management.**
- **Profile Screen: Developed the user profile screen with account information and related features.**

George Joseph (UX & API Developer):

- **API Integration: Implemented the Google Books API integration and network data flow.**
- **Home Screen: Developed the main discovery interface (Banners & Book Lists).**
- **Data Presentation: Connected API responses with the UI and managed state updates.**

Zainab Ahmed (Frontend Developer):

- **Explore Screen: Developed the Explore screen for browsing books.**
- **Category Filtering: Implemented category-based filtering for book discovery.**
- **Search Functionality: Developed the search feature to find books efficiently.**
- **Visual Identity: Created the Splash Screen and ensured Material 3 design consistency.**

Mohamed Zakaria (Book Details Developer):

- **Book Details Screen: Developed the detailed view for displaying comprehensive book information.**
- **Book Actions: Implemented the Buy and Read Now actions for user interaction.**
- **UI Integration: Connected book data with the details screen and ensured a smooth user experience.**

Ali Ahmed (Room Database Developer):

- **Room Database: Implemented the local database using Room.**
- **Favorites System: Developed the Favorites feature for saving and managing books.**
- **Data Persistence: Managed local storage and retrieval of user data.**

1.4 Risk Assessment & Mitigation Plan

This section identifies potential risks that could impact the project's timeline or quality and outlines the strategies to mitigate them.

Risk Category	Potential Risk	Mitigation Strategy (Solution)
Technical	Slow performance when loading large PDF files or long book lists.	Implement **Pagination** for API calls and use **Asynchronous Loading** with Coroutines to keep the UI responsive.
Data Security	Risk of losing user "Favorites" or local data if the app is cleared.	Use **Room Database** for persistent local storage, ensuring data remains available offline and across sessions.
External	Changes or downtime in the external Book API.	Implement a **Repository Pattern** with local caching so the app can display previously fetched data even if the API is offline.
Project Management	Tight deadlines for developing multiple complex features (Search, PDF Reader, etc.).	Adopt an **Agile approach** by focusing on the **Minimum Viable Product (MVP)** first, ensuring core features are stable before polishing extras.

Risk Category	Potential Risk	Mitigation Strategy (Solution)
Integration	Conflicts when merging code from 5 different team members on GitHub.	Enforce strict **Branching Strategies** and perform regular code reviews to ensure all modules integrate seamlessly without breaking the app.

1.5 KPIs (Key Performance Indicators)

To ensure the success and quality of the **BookNest** application, we have defined the following metrics to measure performance and user satisfaction:

- **System Stability:** Achieve a **high crash-free rate** to ensure a smooth and reliable reading experience for all users.
- **Performance & Speed:** Book metadata and covers should load within **1.5 to 2 seconds** using optimized API calls and image caching.
- **Data Integrity:** Ensure **100% accuracy** in syncing the "Favorites" list between the UI and the local Room Database.
- **User Engagement:** A successful test run where a user can search, find, and open a PDF book in **less than 4 clicks**.
- **Resource Efficiency:** Maintain low memory and battery consumption by optimizing image loading (Coil/Glide) and using efficient background threads (Coroutines).

2. Literature Review

Introduction

Before developing BookNest, it is important to study existing e-book platforms and digital libraries to understand their strengths and weaknesses. This helps improve our system and provides better user experience.

Similar Systems:

1. Amazon Kindle

A popular platform for reading and purchasing e-books online.

Features:

- Huge collection of books
- Paid and free books
- Reading on multiple devices
- Bookmarking and notes

Weaknesses:

- Some books are expensive
- Requires an Amazon account
- Interface may be complex for some users

2. Google Play Books

Online service for buying and reading books.

Features:

- Easy login with Google account
- Cloud synchronization
- Free samples available

Weaknesses:

- Limited Arabic books
- Some books are unavailable in some regions

3. Goodreads

Platform for discovering books and reading reviews.

Features:

- Book ratings and reviews
- Recommendations
- Reading lists

Weaknesses:

- Not focused on direct reading
- Interface crowded sometimes

Comparison Table

System	Free Books	Paid Books	Download	Reviews
Kindle	Yes	Yes	Yes	Yes
Google Play Books	Yes	Yes	Yes	Limited
Goodreads	No	No	No	Yes
BookNest	Yes	Yes	Planned	Planned

Suggested Improvements for BookNest

BookNest can be improved by adding:

- Clean and simple user interface
- Arabic and translated books
- Fast search system
- Book reviews and ratings
- Recommendation based on the user’s activity
- In-app book purchase & preview

Conclusion

After analyzing similar systems, BookNest aims to combine the best features of existing platforms while solving common problems to create a better digital library experience.

3.Requirements Gathering

Overview:

BookNest is a mobile application that leverages the Google Books API to provide users with access to a vast library of books. The app categorizes books into Free, Paid, and Translated collections, allowing users to browse, preview, purchase, and save their favorite titles. With an intuitive interface and seamless integration with Google Books, BookNest aims to enhance the digital reading experience.

Objectives

- Provide easy access to Google Books API library
- Enable users to browse books by categories (Free, Paid, Translated, etc...)
- Enable users to search books by name
- Implement a favorites system for personalized collections
- Deliver a smooth and intuitive user experience
- Ensure secure authentication and data protection

Users

Target Audience: General readers of all ages who enjoy digital reading, including students, professionals, and book enthusiasts seeking convenient access to a wide range of books.

Functional Requirements

Authentication

- Sign Up: Users can create accounts with email and password
- Login: Secure authentication for returning users
- Logout: Safe session termination

Browse & Categories

- Display books in three main categories: Free, Paid, Translated
- Smooth scrolling and navigation between categories
- Thumbnail previews with book covers

Book Details

- Display comprehensive book information (title, author, description, publisher, etc.…)
- Show availability status and pricing
- Display book cover in high quality

Book Actions (Coming Soon)

- Preview: Access sample pages or preview links
- Download: Download available free books
- Buy: Redirect to purchase links for paid books

Favorites

- Add books to favorites list
- Remove books from favorites
- View all saved favorites in dedicated section

Search

- Search books by title, author, or keywords
- Real-time search results
- Filter and sort options

Non-Functional Requirements

Performance:

- Fast loading times (under 3 seconds)
- Smooth scrolling and transitions
- Efficient API calls and caching

Usability:

- Intuitive and user-friendly interface
- Clear navigation and visual hierarchy
- Responsive design for various screen sizes

Security:

- Encrypted user authentication
- Privacy protection for user information

Compatibility:

- Compatible with multiple device sizes
- Minimum OS requirements clearly defined

Reliability:

- 99% uptime availability
- Error handling and user feedback
- Stable performance under various conditions

Use Cases

Use Case 1: Browse and Preview a Book

1. User opens the app and logs in
2. User navigates to a category (e.g., Science)
3. User scrolls through available books
4. User taps on a book to view details
5. User clicks Preview to read sample pages (Planned)
6. User decides to add book to Favorites

Use Case 2: Search for a Specific Book

1. User taps on the Search icon
2. User enters book title or author name
3. System displays matching results
4. User selects desired book from results
5. User views book details and takes action

Use Case 3: Manage Favorites

1. User navigates to Favorites section
2. System displays all saved books
3. User can view details or remove books
4. User can access favorites offline

Suggested Screens

• **Splash Screen:** App logo with loading animation
• **Login / Sign Up:** Clean authentication forms
• **Home:** Category tabs with book grid layout
• **Book Details:** Full book information with action buttons
• **Favorites:** Grid view of saved books
• **Search:** Search bar with results display

Future Enhancements

• **Dark Mode:** Toggle between light and dark themes

- **Reviews & Ratings:** User-generated book reviews and ratings
- **Recommendations:** AI-powered personalized book suggestions
- **Offline Reading:** Download and read books without internet
- **Social Sharing:** Share favorite books with friends
- **Reading Progress Tracker:** Monitor reading habits and goals
- **In-App Purchase & Preview :** Implemented book purchase and preview functionality.

Summary

This document details the requirements for the BookNest mobile application, covering its objectives, user base, functional and non-functional specifications, use cases, suggested screens, and potential future enhancements. It serves as a guide for the development team to create a user-friendly and feature-rich application for accessing and enjoying books.